Seeding Markov Chain Monte Carlo algorithms

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Abstract

The purpose of this work is to discuss and illustrate the need of seeding Markov Chain Monte Carlo algorithms when dealing with unobserved variables, as well as to propose a seeding procedure that relies on a distance-based approach for such cases.

1 Introduction

In real-world scenarios, we often aim to explain a set of traits of a phenomenon of interest under specific conditions. From a probabilistic perspective, we usually propose a stochastic mechanism to approximate the underlying law that is governing the phenomenon of interest in such a way that the outcomes of such a stochastic mechanism can be regarded as realisations of the set of traits, say $\pi(\mathbf{z}|\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is an unobservable random vector that fully defines the proposed stochastic mechanism.

The aforementioned proposal is the core of Statistics, a probabilistic framework for drawing statements about the set of traits by leveraging available measurements $\mathbf{z}_1$, $\mathbf{z}_2$, ..., $\mathbf{z}_N$ of them, here denoted by $\mathbf{z}^{obs}$. In this context, there are two common approaches. On the one hand, the frequentist approach focuses on estimating $\boldsymbol{\theta}$, say $\widehat{\boldsymbol{\theta}}$. On the other hand, the Bayesian approach (Bernardo and Smith, 1994) requires the elicitation of $\pi(\boldsymbol{\theta})$, also known as the prior distribution, but returns (via Bayes' rule) $\pi(\boldsymbol{\theta}|\mathbf{z}^{obs})$, also known as the posterior distribution.

[IN THIS PARAGRAPH WE SHOULD JUSTIFY OUR (BAYESIAN) APPROACH.]

Except for very simple cases, computing and working with the posterior distribution requires numerical methods. These methods are mostly needed to compute expected values with respect to the posterior distribution, so the main problem is computing (or approximating) integrals. Traditional analytical methods, such as the Laplace approximation [REFERENCE] and classical numerical methods, such as quadrature [REFERENCE], have proved to be adequate in a wide range of problems, but do not necessarily scale well. For high dimensional posterior distributions, Markon chain Monte Carlo (MCMC) methods are usually preferred [REFERENCE]. MCMC also makes it possible to deal with latent (unobservable) variables, random effects and missing data.

In the next section, we provide a brief overview of the current state-of-the-art MCMC software. We then address the problem of seeding MCMC algorithms when dealing with unobserved variables, and describe the proposed seeding algorithm for overcoming such scenarios in Section 3, whereas in Section 4 we illustrate it with a case study of Ecology using the R packages **NIMBLE** (de Valpine et al., 2026) and **cmdstanr** (Gabry et al., 2025). Finally, in Section 5 we present some concluding remarks.

2 MCMC in practice

Markov Chain Monte Carlo (MCMC) methods (Brooks et al., 2011) are widely used in practice for reconstructing posterior distributions. These methods rely on the theoretical principle of defining a homogeneous, Harris-recurrent, and aperiodic Markov chain whose stationary probability density function is the target posterior distribution; that is, to construct a suitable Markov chain $\tilde{\boldsymbol{\theta}}^{(1)}, \tilde{\boldsymbol{\theta}}^{(2)}, \ldots$ such that $\tilde{\boldsymbol{\theta}}^{(t)} \underset{t\to\infty}{\sim} \pi(\boldsymbol{\theta}|\mathbf{z}_{1:N})$. See Meyn and Tweedie (2009) for a concise treatment of the theory of Markov chains.

Historically, the BUGS project (Lunn et al., 2012) led the efforts to enable Bayesian methods and JAGS (Plummer, 2003) as one BUGS engine.

Lately, Hamiltonian Monte Carlo (HMC) methods (Neal, 2011) have became an alternative to MCMC methods.

HMC is the workhorse of the Stan software (Stan Development Team, 2026). On the other hand, the NIMBLE system (de Valpine et al., 2017).

3 Seeding MCMC algorithms

3.1 Motivation

A more challenging scenario is when, despite having a well-defined stochastic mechanism, the set of traits is comprised of unobservable and observable variables, that is, if $\mathbf{z} = (\mathbf{x}, \mathbf{y})$ we only have measurements for $\mathbf{y}$, say $\mathbf{y}^{obs}$. At first sight this might seem far-fetched, however, it is precisely the setting of state-space models, where the state vector is unobservable and the output vector is observable. In this case, we instead seek to reconstruct $\pi(\mathbf{x}, \boldsymbol{\theta}|\mathbf{y}^{obs})$, and thus, we have not only initialise $\boldsymbol{\theta}$ but also $\mathbf{x}$. Built upon a distance-based approach, we proposed the following seeding process for MCMC algorithms when dealing with unobserved variables.

3.2 Algorithm

Given a prior distribution for $\boldsymbol{\theta}$, $\pi(\boldsymbol{\theta})$, we start by drawing a sample from it, say $\boldsymbol{\theta}^{(t)}$. Next, we simulate a realisation of $\pi(\mathbf{x}, \mathbf{y}|\boldsymbol{\theta}^{(t)})$, say $(\mathbf{x}^{(t)}, \mathbf{y}^{(t)})$, which can be regarded as a candidate measurement of the full set of traits $\mathbf{z}$. Now, let $d : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ be a distance function and d^{max} a distance threshold. We can argument that the candidate measurement $\mathbf{y}^{(t)}$ is at the bulk of $\pi(\mathbf{y})$ if $d(\mathbf{y}^{(t)}, \mathbf{y}^{obs}) < d^{max}$. In that case, we can also argument that the candidate measurement $\mathbf{x}^{(t)}$ is at the bulk of $\pi(\mathbf{x})$. We thus have a potential initial value for the MCMC algorithm that can help us with reconstructing $\pi(\mathbf{x}, \boldsymbol{\theta}|\mathbf{y}^{obs})$, namely, $(\boldsymbol{\theta}^{(t)}, \mathbf{x}^{(t)})$. Algorithm 1 summaries such a seeding process.

Algorithm 1 Distance-based seeding process

Require: (Set) d^{max}
Require: $\boldsymbol{\theta}^{(t)}$
Ensure: $\boldsymbol{\theta}^{(t)}, \mathbf{x}^{(t)}$
1: $t = 0$
2: **repeat**
3: $\quad t \leftarrow t + 1$
4: $\quad \boldsymbol{\theta}^{(t)} \sim \pi(\boldsymbol{\theta})$
5: $\quad \mathbf{x}^{(t)}, \mathbf{y}^{(t)} \sim \pi(\mathbf{x}, \mathbf{y}|\boldsymbol{\theta}^{(t)})$
6: **until** $d(\mathbf{y}^{(t)}, \mathbf{y}^{obs}) < d^{max}$
return $\boldsymbol{\theta}^{(t)}, \mathbf{x}^{(t)}$

3.3 Discussion

The proposed seeding process relies on two ideas: (i) to draw samples from the prior predictive distribution for simulating realisations of $(\mathbf{x}, \boldsymbol{\theta})$, and (ii) to use the results of Maa et al. (1996) for assessing when those simulated realisations are somehow close to the mode of $\pi(\mathbf{y}^{obs}|\mathbf{x}, \boldsymbol{\theta})$.

4 Case study: Population with exponential growth

As we have mentioned above, in state-space models there are both unobserved and observed variables. We thus illustrate the need of seeding MCMC algorithms when dealing with unobserved variables with a case study of Ecology in which the proposed stochastic mechanism is formulated as a state-space model, namely population growth with an exponential rate. The implementation was carried out in R (R Core Team, 2026) and for reproducibility purposes the code is available at `https://github.com/roman-ap/SeedingStan`.

Auger-Méthé et al. (2021) outline how state-space models can be adopted for modelling ecological time series, including the advantages and disadvantages of using a Bayesian approach. On the other hand, Monnahan et al. (2017) propose to use HMC for mitigating the computational burden of traditional MCMC algorithms when dealing with complex Bayesian models for ecological data. We will use both traditional MCMC and HMC algorithms with the assistance of the R packages **NIMBLE** (de Valpine et al., 2026) and **cmdstanr** (Gabry et al., 2025), respectively.

4.1 Model

For a discrete-time setting, let us assume that a population experience an exponential growth with time-varying rate r_t, that is, $N_{t+1} = e^{r_t} N_t$ for $t = 1, 2, \ldots, T-1$. We can rewrite the time-varying rate as $r_t = f_t - m_t$, where $f_t, m_t \in \mathbb{R}^+$ are the fertility and mortality rates at time t, respectively. Now, at each time t, we can think of abundance before and after mortality, say N_t^{pre} and N_t^{post}, respectively. Thus, if M_t denotes the total mortality we have that $N_t^{post} = N_t^{pre} - M_t$.

Given the aforementioned population demographics, we can propose the following stochastic population mechanism:

$$
\begin{aligned}
M_t &\sim \text{Bin}\left(N_t^{pre}, 1 - e^{-m_t}\right) \\
N_{t+1}^{pre} &\sim \text{Poi}\left(e^{f_t} N_t^{post}\right)
\end{aligned}
$$

Furthermore,

$$
\begin{aligned}
\log(N_1^{pre}) &\sim \text{Poi}\left(e^{LN_0}\right) \\
\log(a_t) &\sim \text{N}\left(\mu_{la}, \sigma_{la}\right) \\
\log(b_t) &\sim \text{N}\left(\mu_{lb}, \sigma_{lb}\right) \\
\log(f_t) &\sim \text{N}\left(\mu_{lf}, \sigma_{lf}\right)
\end{aligned}
$$

with

$$
\begin{aligned}
\mu_{lm} &\sim \text{N}\left(m_{lm}, sd_{lm}\right) \\
\sigma_{lm} &\sim \text{N}\left(0, sd_{lm}\right) \\
\mu_{lf} &\sim \text{N}\left(m_{lf}, sd_{lf}\right) \\
\sigma_{lf} &\sim \text{N}\left(0, sd_{lf}\right)
\end{aligned}
$$

4.2 Simulated data

4.3 Results

5 Concluding remarks

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